

Anti-transforming growth factor- β treatment shows increased bone mass and strength in a novel mouse model for osteogenesis imperfecta type I

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Abstract

Anti-transforming growth factor beta (TGF- β) is a promising approach for the treatment of osteogenesis imperfecta (OI). To date, preclinical and clinical studies for the use of anti-TGF- β therapy have focused on moderate to severe OI caused by qualitative defects in collagen. However, the majority of OI patients are represented by type I OI. Mutations resulting in the haploinsufficiency of type I collagen is the cause of OI type I in the majority of patients. To study the effect of anti-TGF- β therapy in type I OI, we generated a novel mouse model for OI type I. CMV-CRE mice were crossed to mice where *Col1a1* was floxed between exon 2 and 5 to create a full body heterozygous deletion of *Col1a1*. Haploinsufficiency of *Col1a1* in the tibia was confirmed by decreased *Col1a1* mRNA and protein expression. Comparable to OI patients, we observed reduced bone mass by μ CT in these *Col1a1*^{+/-} mice. Biomechanical measurements showed a decrease in bone strength and an increase in bone brittleness. Histomorphometric analysis showed an increase in osteoclast number and a trend towards increased osteoblast number supporting a high bone turnover phenotype, similar to OI type I patients. Upon treatment with a pan anti-TGF- β antibody, 1D11, *Col1a1*^{+/-} mice showed increased bone mass and improved ultimate strength, but measures of ductility did not show improvement. Overall, our findings support expanding the study of anti-TGF- β treatment to OI caused by haploinsufficiency of type I collagen.

Keywords: osteogenesis imperfecta, transforming growth factor β (TGF- β), collagen type I, OI type I, 1D11

Lay Summary

Osteogenesis imperfecta is a genetic disease associated with skeletal fragility. Here, we show that a mouse model for the most common form of osteogenesis imperfecta, type I, has a bone phenotype that recapitulates patient phenotypes. Treatment with anti-transforming growth factor - β therapy increases bone mass and strength in these mice.

Introduction

Osteogenesis imperfecta (OI) is a rare genetic skeletal disorder with an estimated prevalence of 1 in 15 000–20 000 live births. OI is characterized by fragile bones with frequent fractures and bone deformities and extraskeletal manifestations such as hearing loss and lung abnormalities.^{1–3} Clinically, OI can be classified based on the Sillence classification: OI type I describes a milder form of OI with blue sclera and low bone mass without bone deformities, severe perinatal lethal OI is classified as OI type II, and type III and IV describe moderate to severe OI that includes progressive bone deformities. OI type I is the most common form of OI and represents over 50% of patients.^{4,5} The vast majority of OI (85%–90%) is caused by mutations in either *COL1A1* or *COL1A2* that encode the 2 pro- α chains that compose heterotrimer type 1 collagen. Qualitative mutations that affect collagen

structure, for example, glycine substitutions, lead to delayed collagen triple helix assembly and post-translational over-modification, result in severe forms of the disease. In contrast, quantitative defects as a result of mutations leading to haploinsufficiency of either *COL1A1* or *COL1A2* lead to reduced amount of type I collagen and typically cause OI type I.^{3,6–8} Type I OI patients are slightly shorter with a slight reduction in body weight. While on the milder end of the spectrum, vertebral and long bone fractures occur at a much higher rate in type I OI patients than in the general population.⁶

Currently, there are no U.S. Food and Drug Administration (FDA) approved therapies for OI. Existing treatment options rely on the repurposing of treatments used to treat postmenopausal osteoporosis with anti-resorptive drugs, mainly bisphosphonates, as the current standard of care. Additionally, anabolic therapies are currently being studied, including treatments that target the Wnt pathway such as anti-sclerostin and

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anti-DKK1 antibodies.^{9,10} Reducing transforming growth factor beta (TGF- β) signaling has been shown to increase bone mass in mice¹¹ and excessive TGF- β signaling plays a driving role in OI pathogenesis in multiple models of OI caused by qualitative mutations in collagen type 1. Reduction of this excessive TGF- β signaling enhances bone volume and strength in these models.^{12–14} In addition, patients with moderate to severe OI show increased TGF- β signaling and anti-TGF- β treatment is well tolerated.¹⁵ Therefore, it is important to determine if OI caused by quantitative defects in collagen also responds to anti-TGF- β therapy.

To date, 2 mouse models for haploinsufficient OI type I have been reported, the *Col1a1*^{+/-365} mouse, in which exon 2-5 of *Col1a1* was removed using CRISPR/CAS9¹⁶ and the Mov13 mouse. The Mov13 mouse has an insertion of Moloney murine leukemia provirus (M-MuLV) into the first intron of the *Col1a1* gene. Mice with homozygous insertion are embryonically lethal, but the heterozygous Mov13 mice have reduced *Col1a1* transcription and was described as a model of OI type I.^{17,18} However, this mouse model has several caveats; while expression of the *Col1a1* allele is blocked in fibroblasts, it is expressed, albeit in reduced manner, by osteoblasts,¹⁹ bone strength and mineralization returned to normal by 16 wk¹⁸ and these mice show early development of lymphoma.²⁰

To study the effect of anti-TGF- β treatment on quantitative OI, we generated a novel *Col1a1*^{+/-} mouse model which recapitulated patient phenotypes such as reduced collagen expression, decreased bone mass, and reduced bone strength. We treated this mouse model with 1D11, a pan TGF- β antibody, and observed increased bone mass and biomechanical properties. This treatment outcome suggests that patients with quantitative defects in type I collagen may benefit from anti-TGF- β therapy.

Materials and methods

Mice

All mouse experiments were performed according to a protocol approved by the Institutional Animal Care and Use Committee at Baylor College of Medicine. Sperm from *Col1a1*^{loxP/LoxP} mice with loxp flanking between exon 2 and 5 were a kind gift from Dr Raghu Kalluri.²¹ These mice were crossed to CMV-Cre to cause heterozygous loss of *Col1a1*. Mice were then crossed to C57BL/6J mice to eliminate the Cre allele. Mice were housed in groups with unlimited access to water and food. Their diet was Picodiet Select rodent 5V5R diet 50 IG/6F (extruded) procured from Labdiet.

1D11 treatment

Mice were randomly assigned to be treated with mouse anti-TGF- β antibody 1D11 or 13C4 control antibody. Ten mice per group were assigned to acquire appropriate power and accommodate potential loss of samples. Final group size was between 8 and 11 as sometimes mice or samples were lost due to unexpected events. Antibodies were kindly provided by Sanofi. Treatment started at 8 wk of age; IP injections were performed once weekly at 10 mg/kg for a duration of 8 wk. 1D11 or 13C4 treated mice were euthanized at 16 wk old and body weight was recorded at time of injection and at euthanasia. Untreated mice were euthanized at 12 wk of age and body weight was recorded at time of euthanasia. Calcein was injected at a 7-d interval, 9 and 2 d prior to euthanasia.

Left femurs were wrapped in PBS-soaked gauze and stored at -20°C for μCT measurement followed by 3 point bending analysis. Bone marrow was flushed from tibias and they were flash frozen and stored at -80°C . Spine was fixed in 4% paraformaldehyde for 48 hr and stored in 70% ethanol prior to μCT and plastic embedding. Investigators were blinded during μCT , histomorphometry and 3-point-bending analysis.

RNA

Flash frozen tibias were used for RNA extraction using Direct-Zol kit (Zymo research #R2052). Reverse transcription was performed using Superscript IV (ThermoFisher Scientific #18090010). quantitative reverse transcription polymerase chain reaction (q-RT-PCR) was performed for *Col1a1* and normalized to β -actin expression (Table S1).

For RNA sequencing, isolated RNA was sent to Azenta Life Sciences for poly-A selection and library preparation and sequencing. The quality of the sequencing reads was first assessed with FastQC (Version 0.11.8). We then used Trimmomatic (Version 0.39) to remove low quality reads and adapter sequences. All reads were aligned to the mouse genome version GRM39 using Hisat2 (Version 2.2.1). The Samtools package (Version 1.20) was used to convert, sort and index the reads. Lastly, Featurecounts from the subread package (Version 2.0.3) was used to generate counts of all mapped reads. Deseq2 (Version 1.42.1) was used to identify differentially expressed genes. ShinyGO (V0.81) was used for gene ontology (GO) and Kyoto encyclopedia of genes and genomes (KEGG) analysis.^{22–24}

Western blot analysis

Protein was extracted from tibia using Radio-Immunoprecipitation Assay (RIPA) buffer and quantified using Pierce BCA protein assay kit (ThermoFisher Scientific #23225). Ninety micrograms of protein was loaded onto the SDS-PAGE. Pro-Col1a1 (Sigma Millipore #ABT257) was used at 1:500. Anti-Vinculin (Santa Cruz sc73614) was used at 1:1000 as a loading control. Densitometry was performed using Image Lab software (Biorad).

μCT

μCT was performed using the Scanco μCT -40 system (Scanco Medical AG, 55-peak kilovoltage, and 145- μA X-ray source) at 16 μm resolution. Trabecular bone of the L4 vertebrae and distal femoral metaphysis were analyzed. Trabecular analysis of the femur was measured by analyzing 75 slides in the trabecular region of the distal femur. Trabecular analysis of the L4 vertebrae was measured between the L4 vertebral endplates. Femoral cortical analysis was performed by analyzing 11 slides at the femur midshaft.

Histomorphometry

The undecalcified lumbar vertebra 4 and 3 specimens were dehydrated in increasing concentrations of ethanol and embedded in methylmethacrylate.²⁵ About 4.5- μm -thick sections were cut with a tungsten carbide blade (Delaware Diamond Knives, Inc.). The sections were deplastified with 2-methoxyethyl acetate. For each sample Goldner Trichrome, Von Kossa, and tartarate resistant acid phosphatase staining was performed. One unstained section was used for the measurement of dynamic parameters using fluorescence microscopy. Results were quantified using Bioquant Osteo 2023 (V23.5.60) software.

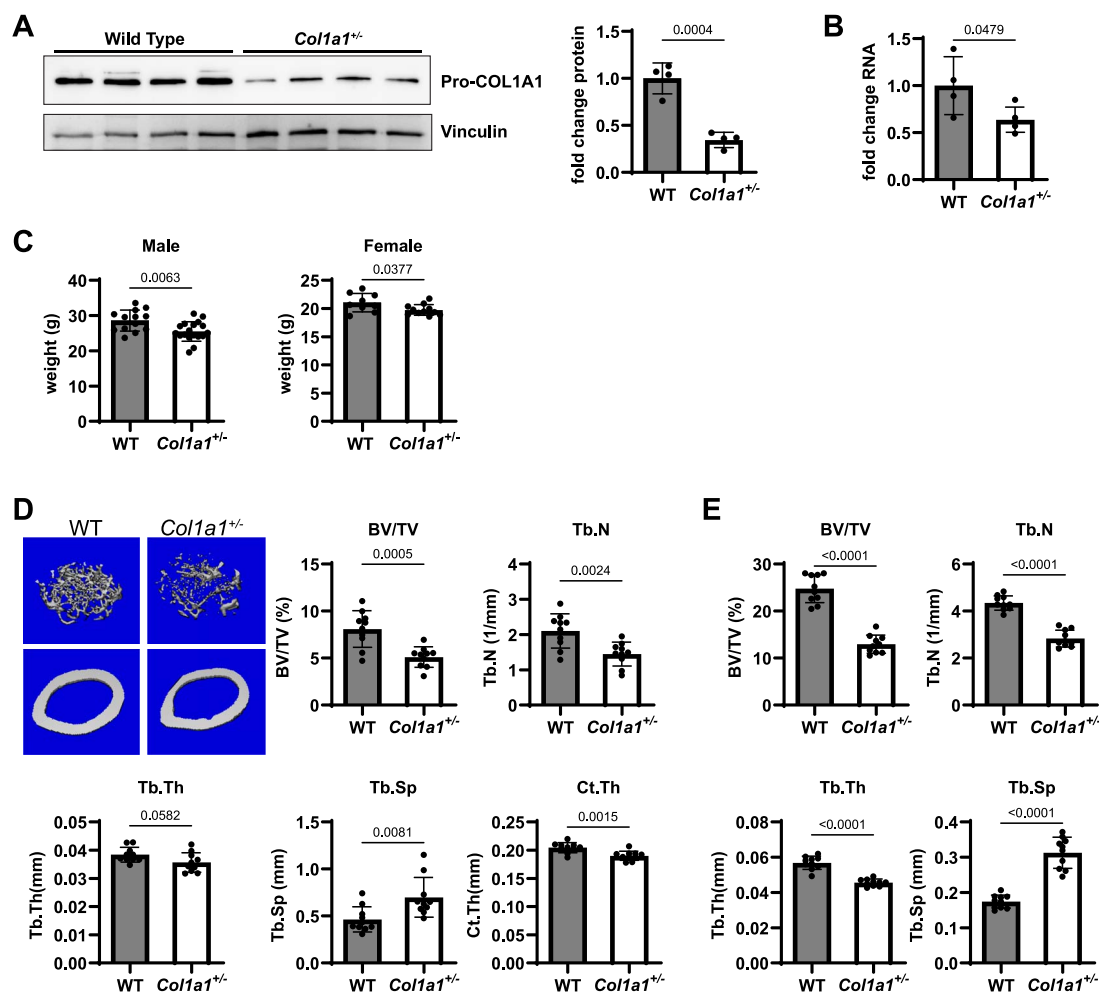


Figure 1. Heterozygous deletion of exon 2-5 of *Col1a1* gene results in decreased bone mass. (A) Western blot analysis of tibias of 12-wk-old male WT and *Col1a1*^{+/-} mice showing pro-collagen type I, alpha 1 relative to vinculin. Quantification is shown on the right. (B) q-RT-PCR analysis of *Col1a1* mRNA in RNA extracted from tibias of 12-wk-old male WT (*n*=4) and *Col1a1*^{+/-} (*N*=5) mice. (C) Weights (g) of male (left) and female (right) WT and *Col1a1*^{+/-} mice at 12 wk. (Male WT *N*=13, *Col1a1*^{+/-} *N*=18, female WT *N*=9, *Col1a1*^{+/-} *N*=10). (D) μCT analysis results of left femurs of 12-wk-old female mice (*n*=10). Left: representative images of trabecular and cortical bone. Right: bone volume/total volume (TV/TV), trabecular number (Tb.N), trabecular thickness (Tb.Th), trabecular separation (Tb.Sp) and cortical thickness (Ct.Th) of WT and *Col1a1*^{+/-} mice are shown. (E) μCT analysis results of the L4 vertebrae from female 12-wk-old WT and *Col1a1*^{+/-} mice (*n*=10). Results are shown as means ± SD. *p* values were determined using student's *t*-test.

Three-point-bending

Thawed, rehydrated femurs were tested to failure in 3-point bending at a rate of 0.1 mm/s and were oriented in the test fixture such that the anterior surface was in compression and the posterior surface in tension. Femurs were tested wet at room temperature using a 100 N load cell on an Instron 6800 series single column table model (68SC-2) (Instron Inc.). Load and displacement data were captured at a rate of 50 Hz using Instron's Bluehill software.

The maximum load was determined by finding the highest load value recorded before the specimen fractured. The region of the load-displacement curve between 1 N and the maximum load was separated into 5 segments and the fitted line of the segment with greatest slope was defined as the stiffness. A line representing 10% degradation of this stiffness was used to define the yield point. The elastic region was identified as the region from the completion of the preload to the yield point. The post-yield region was identified as the region from the yield point until point of specimen fracture. Using a trapezoidal numerical integration method, the work to fracture was calculated as the area under the load-displacement curve. The cross-sectional geometry of each bone as determined by μCT

image analysis was used to convert the maximum load and stiffness data into ultimate stress and elastic modulus values using beam theory.

Statistics

For comparisons between 2 groups 2 tailed student *t*-test were performed. For comparisons among 4 groups 2-way ANOVA was performed followed by multiple comparisons with Holm correction using Graphpad Prism (V10) software. False discovery rate was used to determine significance in RNA-seq analysis.

Results

To generate a mouse model with *Col1a1* haploinsufficiency we crossed mice with a floxed *Col1a1* allele between exon 2 and 5²¹ to CMV-Cre transgenic mice and subsequently the Cre transgenic allele was eliminated by selective breeding. We tested the expression of *Col1a1* levels and found an expected approximately 50% reduction in protein and RNA level in tibias of heterozygous vs wild type (WT) mice (Figure 1A and B). Compared to littermate controls, *Col1a1*^{+/-} mice

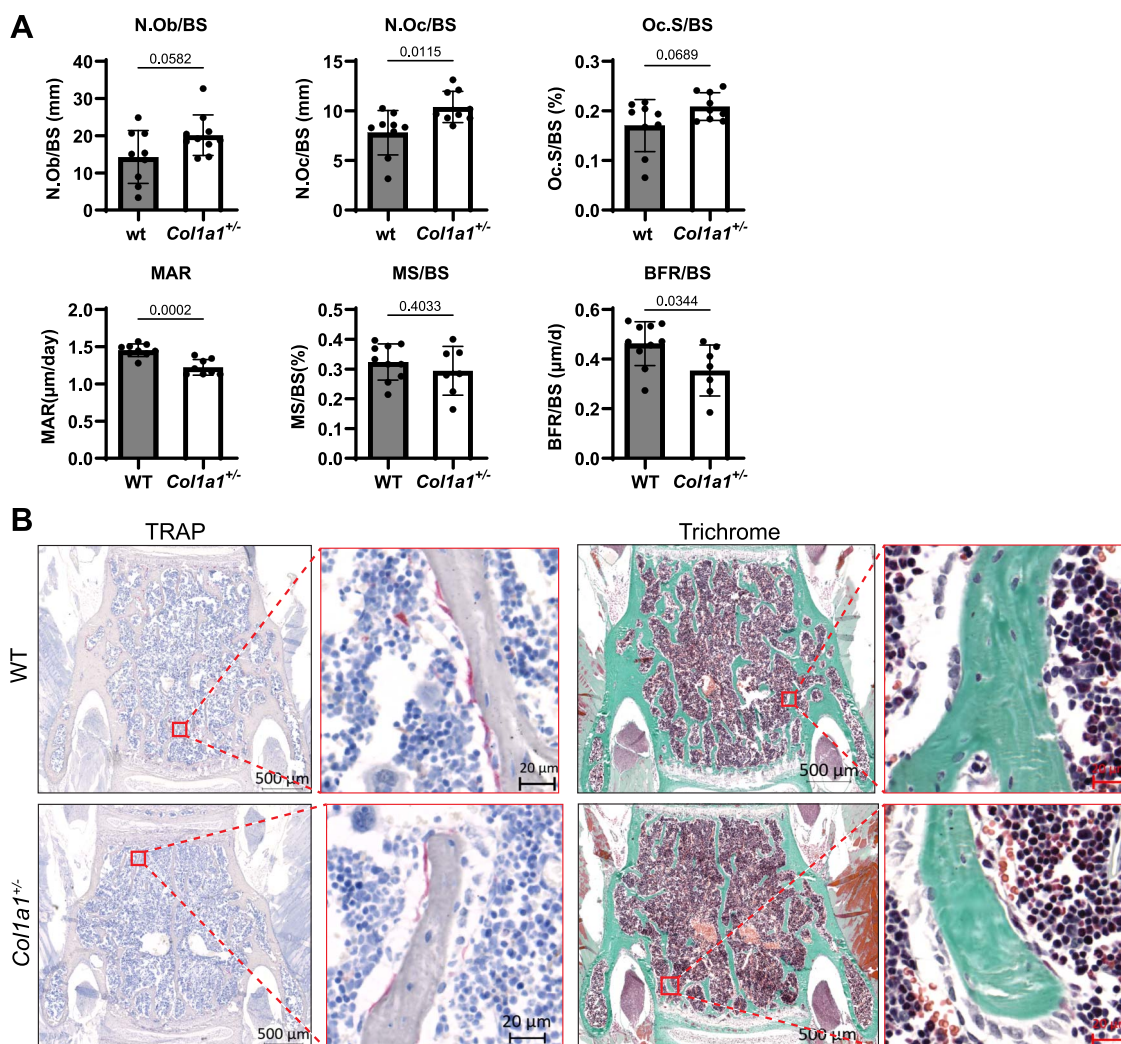


Figure 2. Histomorphometric analysis of WT and *Col1a1*^{+/-} vertebrae. (A) Number of osteoblast per mm bone surface (N.Ob/BS), number of osteoclast per mm bone surface (N.Oc/BS), and osteoclast surface per mm bone surface (Oc.S/BS), mineral apposition rate (MAR), mineralizing surface/one surface % (MS/BS) and bone formation rate (BFR) in 12-wk-old female WT and *Col1a1*^{+/-} mice. Results are shown as means $\pm$ SD. *p* values were determined using student's *t*-test. (B) Representative images of WT and *Col1a1*^{+/-} mouse vertebrae stained with tartarate resistant acid phosphatase (left) and trichrome (right).

have a slight, but significantly reduced body weight at 12 wk (Figure 1C). Both male and female *Col1a1*^{+/-} mice show a reduction in trabecular bone volume (BV/TV), trabecular number (Tb.N) and an increase in trabecular separation (Tb.Sp) in the spine and femur (Figure 1D, E and Figure S1). Additionally, the femurs have reduced cortical thickness (Ct.Th) (Figure 1D and Figure S1). Overall, these data support that *Col1a1*^{+/-} mice demonstrate trabecular and cortical bone phenotypes reminiscent of OI type I patients.

To explore the cellular mechanism for this decrease in bone mass we performed bone histomorphometry on vertebral sections. Consistent with μ CT analysis, we found reduced BV/TV on quantification of von Kossa-stained *Col1a1*^{+/-} spines. We further observed an increase in osteoclast number and a trend towards increased osteoblast number in *Col1a1*^{+/-} animals. Dynamic histomorphometry showed a decrease in mineral apposition rate (MAR) and bone formation rate (BFR) while the amount of mineralizing surface compared to bone surface (MS/BS) was not changed in *Col1a1*^{+/-} animals compared to controls (Figure 2 and Figure S2).

To assess bone strength, we tested the biomechanical properties of the *Col1a1*^{+/-} mice by performing 3-point bending assays on the femurs of 12-wk-old mice. In female mice, we observed a reduction in ultimate load, stiffness, post-yield displacement and total energy (Figure 3A). This was likely due to changes in bone material properties as there was a concurrent reduction in intrinsic measures for ultimate strength, elastic modulus, post-yield strain and toughness to failure (Figure 3B). Male mice showed similar features, except no significant change in elastic modulus was seen (Figure S3). Overall, these data suggest that *Col1a1*^{+/-} OI mice have more brittle bones, that is, they are less resistant to bending and break more readily.

To assess changes in global gene expression, RNA-seq analysis was employed. RNA-seq analysis identified significant changes in 2086 genes. GO analysis of significantly upregulated genes shows expected changes in pathways such as ossification, skeletal system development and extracellular matrix organization. In addition, KEGG analysis of these genes shows changes in Hedgehog, Wnt and TGF- β signaling

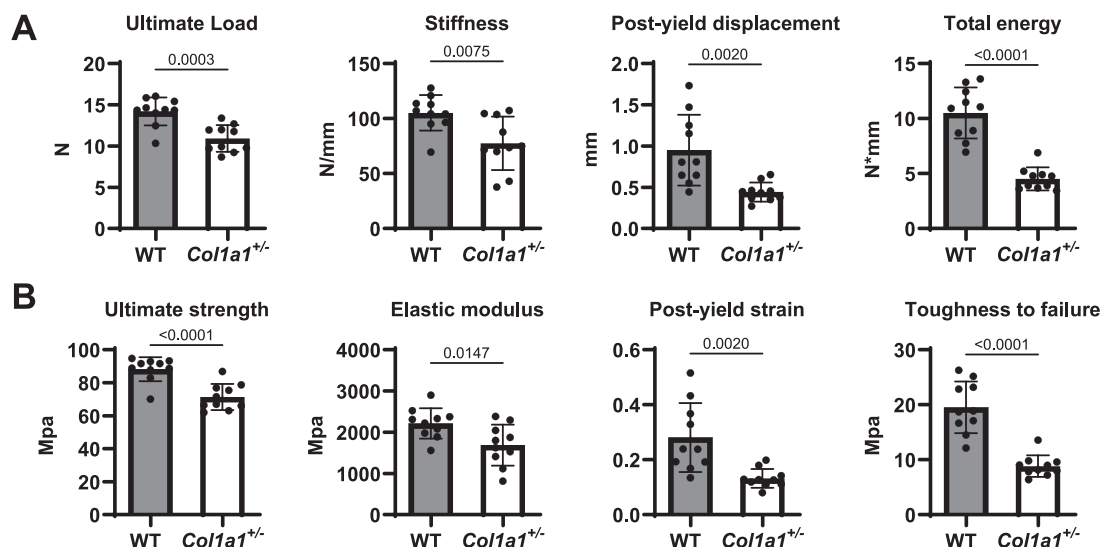


Figure 3. Biomechanical measurements of WT and *Col1a1*^{+/-} mice. Results from 3-point bending analysis of 12-wk-old female WT and *Col1a1*^{+/-} mice ($n=10$). (A) Extrinsic values ultimate load, stiffness, post-yield displacement and total energy. (B) Intrinsic values ultimate strength, elastic modulus, post-yield strain and toughness to failure. Results are shown as means $\pm$ SD. p values were determined using student's t -test.

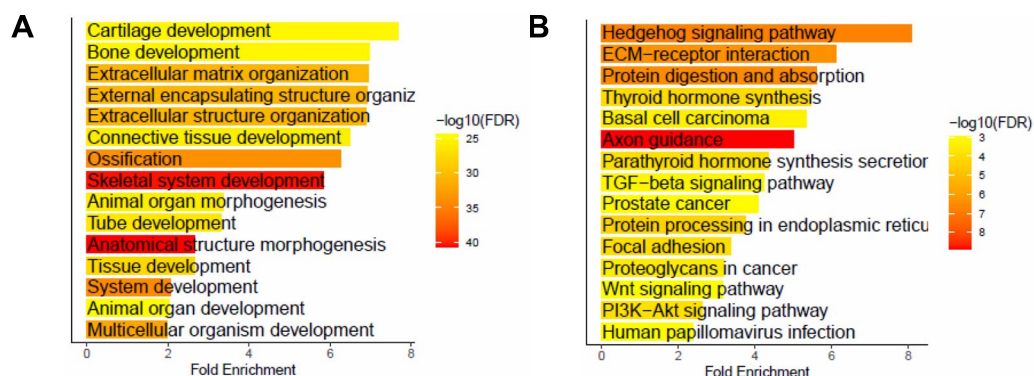


Figure 4. RNA-seq analysis of WT and *Col1a1*^{+/-} mouse tibias show changes in genes enriched for bone related pathways. (A) Top 15 most significantly enriched pathways based on false discover rate (FDR) by gene ontology analysis of significantly upregulated genes (FDR < 0.05). (B) Top 15 most significantly enriched pathways (FDR) identified by KEGG analysis of significantly upregulated genes (FDR < 0.05).

among the top enriched pathways (Figure 4). Downregulated genes mainly show enrichment for metabolism related pathways in the top enriched pathways (Figure S4). This suggests that similar to models of OI caused by qualitative changes in type 1 collagen, TGF- β is among the signaling pathways changed by quantitative changes in *Col1a1*.

We set out to determine if inhibiting TGF- β with TGF- β neutralizing antibody 1D11 has beneficial effects in a model of OI caused by quantitative defects. We treated *Col1a1*^{+/-} and WT littermates once weekly with 10 mg/kg 1D11 or 13C4 control antibody for 8 wk starting at 8 wk of age. 1D11 treatment did not affect the weight of the mice (Figure S5). Three mice died during treatment, 2 of these were treated with 1D11 and 1 was treated with 13C4. Therefore, it is unclear whether this was due to the 1D11 treatment. μ CT analysis showed that 1D11 treatment increased trabecular bone mass in the spine and femurs of both male and female WT and *Col1a1*^{+/-} mice. This is seen by an increase in bone volume, trabecular number, and a decrease in trabecular separation. Female mice showed a larger increase than male mice with a supraphysiological increase in the femur, and only female mice showed increased trabecular thickness. However, cortical thickness was not significantly increased in either sex

(Figure 5 and Figures S6 and S7). Overall, these data show that anti-TGF- β treatment could also increase bone mass in a model of OI type I.

To determine if anti-TGF- β therapy improves bone strength, we assessed biomechanical properties after 8 wk of anti-TGF- β therapy using 3-point bending. Femurs from treated *Col1a1*^{+/-} animals showed an increase in resistance to bending, as is shown by an increase in stiffness and elastic modulus, as well as an increase in ultimate load and ultimate strength. However, post-yield displacement, strain, total energy and toughness to failure were not increased, suggesting that anti-TGF- β treatment does not affect the brittleness of the bones (Figure 6).

Discussion

The current standard of care for children with OI is off-label treatment with bisphosphonates. However, benefit in fracture reduction is inconsistent^{9,26,27} therefore, alternative therapeutic approaches that target the underlying mechanism of OI are needed. Increased TGF- β signaling is a common mechanistic change in OI caused by qualitative changes in collagen.^{12,15} In preclinical studies, treatment with

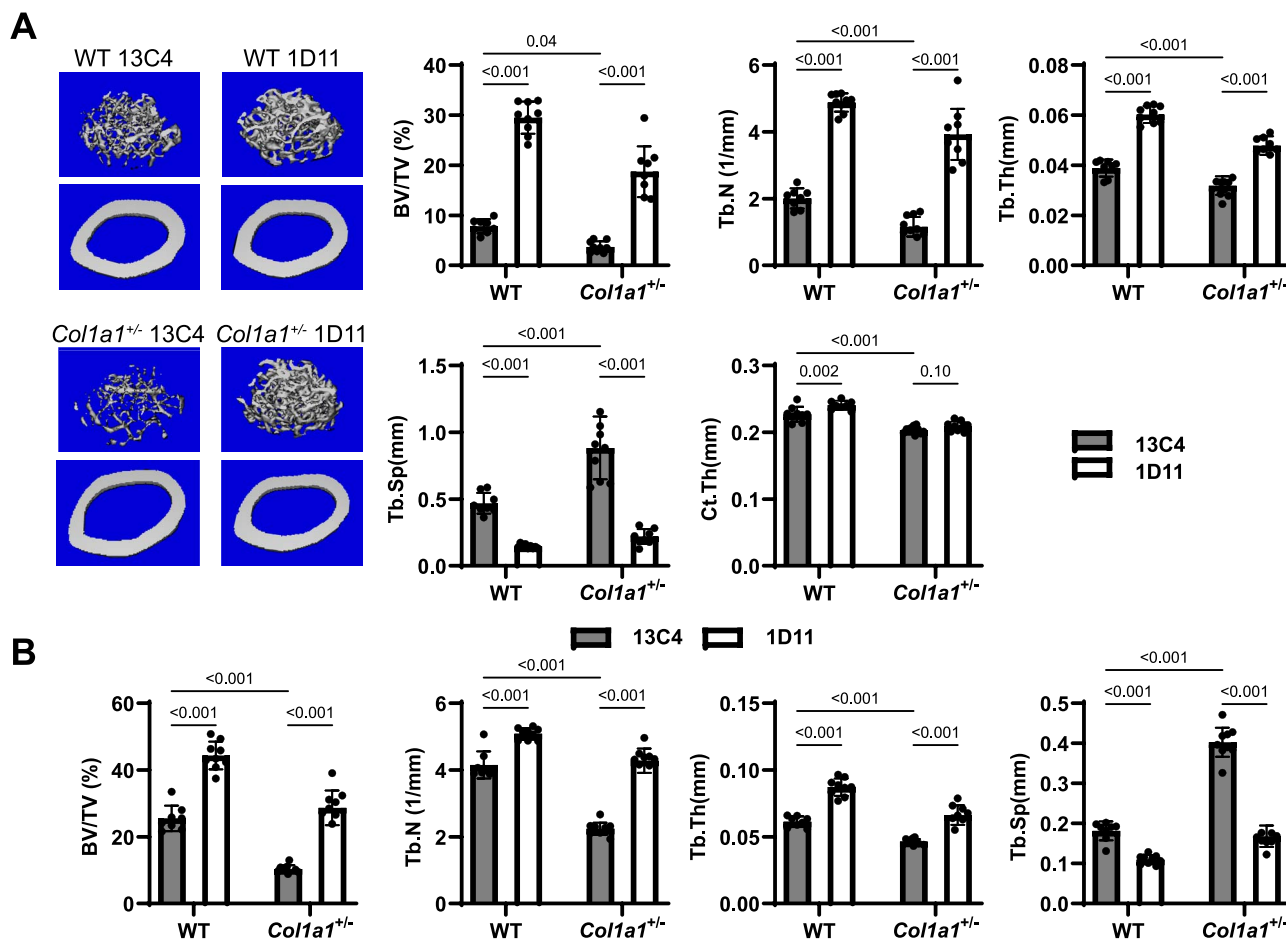


Figure 5. WT and *Col1a1*^{+/-} show increased bone mass mice after treatment with anti-TGF- β antibody 1D11. Eight week-old WT and *Col1a1*^{+/-} mice were treated with weekly IP injections of 10 mg/kg 1D11, or control (13C4) antibody for 8 wk and euthanized at 16-wk-old. (A) μ CT analysis results of left femurs of 16-wk-old female mice. Left: representative images of trabecular and cortical bone. Right: bone volume/total volume (TV/TV), trabecular number (Tb.N), trabecular thickness (Tb.Th), trabecular separation (Tb.Sp) and cortical thickness (Ct.Th). (B) μ CT analysis results of L4 vertebrae of 16-wk-old female WT and *Col1a1*^{+/-} mice. (WT 13C4 N=8, WT 1D11 N=9, *Col1a1*^{+/-} 13C4 N=11, *Col1a1*^{+/-} 1D11 N=10). Results are shown as means $\pm$ SD. *p* values were determined using 2-way ANOVA following multiple comparisons using Holm correction.

anti-TGF- β results in increased bone mass, strength, and a normalization of osteoblast and osteoclast numbers in these OI mouse models.^{12,13} However, preclinical studies to date have focused on models of OI caused by qualitative defects in collagen^{12–14} due in part to lack of suitable mouse models for quantitative collagen type I defects. Because OI type I represents over 50% of total OI patients, there is a significant need to determine the effect of anti-TGF- β therapy in a model of quantitative collagen type I defects. To study the effect of anti-TGF- β treatment for OI type I, we generated a novel OI type I mouse model. This *Col1a1*^{+/-} mouse shows decreased expression of *Col1a1* mRNA and protein in the bone confirming haploinsufficiency mechanism. We further observed osteopenia in both the femur and the spine. The decrease in trabecular bone mass was predominantly due to a decrease in trabecular number, this is consistent with what is seen in OI patients.²⁸ We further show that these mice have more brittle bones with decreased resistance to breaking. On the cellular level, we see a mild relative increase in osteoblast and osteoclast number, and a decrease in MAR. This is also consistent with observations in patients with collagen type 1 mutations.^{28–31} RNA-seq analysis of the *Col1a1*^{+/-} mouse showed significant changes in 2086 genes. Among upregulated genes in RNA-seq analysis, KEGG pathway analysis identified TGF- β signaling in its top

enriched pathways. This suggests that there are changes in TGF- β signaling in these *Col1a1*^{+/-} mice.

We show that mice with *Col1a1* haploinsufficiency benefit from anti-TGF- β therapy. After treatment with 1D11, a pan anti-TGF- β antibody, the femur and spine of female mice show an increase in bone mass with an increase in BV/TV, Tb.N, and Tb.Th with a decrease in Tb.Sp. These findings are similar to what was found in the *Crtap*^{-/-} and *Col1a2*^{+p.G610C} mouse models of OI qualitative defects.^{12–14} We further saw an increase in ultimate load and stiffness in *Col1a1*^{+/-} animals, showing 1D11 treatment made the bone more resistant to bending and increased the breaking load. This was not due to geometric changes as we also saw an increase in ultimate strength and elastic modulus. We did, however, not see any improvement in plastic strain or displacements, suggesting that the bones were still brittle. This is comparable to findings in the *Crtap*^{-/-} and *Col1a2*^{+p.G610C} mouse model where ultimate strength was increased, but plastic and total displacement was not. Interestingly, measures of elastic modulus were increased in the *Col1a1*^{+/-} mice as well as *Col1a2*^{+p.G610C},^{13,14} but not in the *Crtap*^{-/-} mice,¹² and measures of stiffness only appeared to increase in the *Col1a1*^{+/-} mice. For each measure affected by 1D11 in the *Col1a1*^{+/-} mice, we also saw an increase in

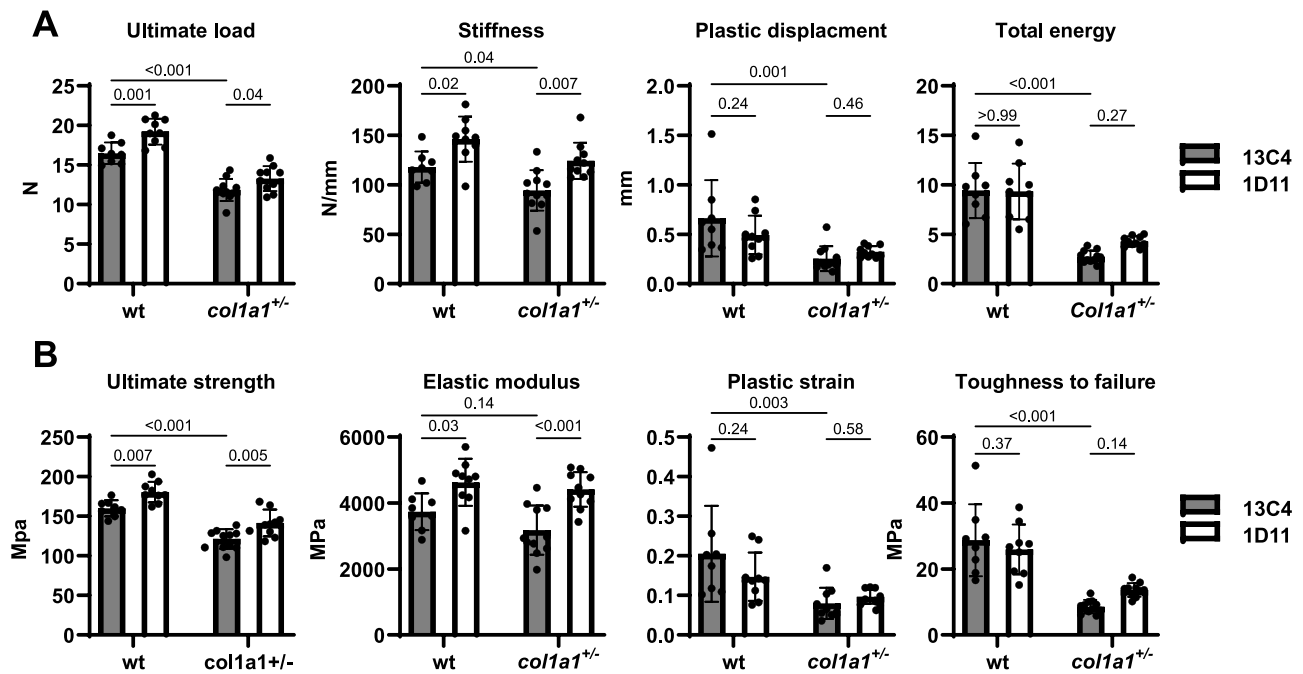


Figure 6. Biomechanical measurements of WT and *Col1a1*^{+/-} mice after control and anti-transforming growth factor- β treatment. Results of 3-point bending analysis of 16-wk-old female WT and *Col1a1*^{+/-} mice that were treated for 8 wk with weekly IP injections of 10 mg/kg 1D11, or control (13C4) antibody. (A) Extrinsic values ultimate load, stiffness, post-yield displacement, and total energy. (B) Intrinsic values ultimate strength, elastic modulus, post-yield strain, and toughness to failure (WT 13C4 $N=8$, WT 1D11 $N=9$, *Col1a1*^{+/-} 13C4 $N=11$, *Col1a1*^{+/-} 1D11 $N=10$). Results are shown as means $\pm$ SD. p values were determined using 2-way ANOVA following multiple comparisons using Holm correction.

the WT littermate controls which is consistent with what was previously reported for WT mice.^{11,32} Interestingly, the most severe *Col1a1*^{tr/+} model of combined OI and Ehlers Danlos syndrome did not improve upon similar doses of TGF- β inhibition.³² Overall, this agrees with the hypothesis that models of milder OI respond better to anti-TGF- β treatment. This could be due to higher dysregulation of TGF- β in more severe forms OI. It is possible that a higher dose of 1D11 is needed in more severe models of OI, however, this likely won't be tolerated well. It has been previously reported that treatment with 1D11 results in decreased bone turnover, with a reduction in BFR, osteoblast and osteoclast number.^{12,13} Therefore, a similar mechanism likely plays a role during TGF- β treatment in the *Col1a1*^{+/-} mice. Importantly, there is a considerable difference between male and female mice in their level of response to anti-TGF- β treatment. During our treatment regimen, female mice showed supra-physiological correction in bone mass, whereas this was not seen in the male femur, and the male spine did not fully correct. This more pronounced change in bone mass in females upon anti-TGF- β treatment was also seen in both WT and *Col1a1*^{+/-} mice. Future studies will need to be performed to understand the mechanism behind this sexual dimorphism.

It has previously been reported that 1D11 treatment in *Col1a1*^{tr/+} mice resulted in death that was likely caused by testicular hemorrhage.³² In our treatment cohort, 3 mice died during treatment. This involved 1 female WT and 1 male *Col1a1*^{+/-} mouse treated with 1D11 and 1 male *Col1a1*^{+/-} mouse that was 13C4 treated. No further analysis was performed on these mice. Since deaths occurred in both groups it is unclear whether these deaths occurred due to 1D11 treatment or unrelated reasons. 1D11 did not impact weight gain during treatment in either group.

Among clinical trials for Fresolimumab, a 1D11 analog, for the treatment of metastatic breast cancer, systemic

sclerosis, steroid-resistant focal segmental glomerulosclerosis, malignant melanoma and renal cell carcinoma, adverse events included keratocanthoma, squamous cell carcinoma, anemia, gastrointestinal and epistaxis and gingival bleeding.^{33–36} Importantly, due to the longer duration of the bone remodeling cycle, anti-TGF- β therapy may not need to be sustained to affect the bone. Therefore, targeting bone with anti-TGF- β therapy may be more feasible. In a phase 1 clinical trial of Fresolimumab for the treatment of OI, no serious adverse events were reported and adverse events included epistaxis, nausea, and blood in urine. Overall, this showed that the anti-TGF- β treatment was well tolerated.¹⁵ Further clinical study is required to determine if anti-TGF- β treatment can be used in OI patients.

Besides enrichment for TGF- β signaling, our RNA-seq analysis identified enrichment for Hedgehog, Pi3k-Akt and Wnt signaling in the *Col1a1*^{+/-} tibia. Other OI treatments currently being investigated include treatment with anti-Sclerostin and anti-DKK1, both of which target the Wnt pathway.^{9,10} It would be interesting to determine the effect of targeting the Wnt pathway in this mouse model.

Previously, 2 mouse models of OI type I have been described, the *Col1a1*^{+/-365} mouse, and the Mov13 mouse. The *Col1a1*^{+/-} mouse model described here differs in many ways from the Mov13 model.^{17,18} The Mov13 mice do not show a change in BV/TV, trabecular number, or separation in the femur and an increase in trabecular thickness and cortical thickness.^{20,37} In the spine, a decrease in bone volume was seen with a reduction in trabecular number.³⁷ This is different in the *Col1a1*^{+/-} model described here, which shows decreased bone volume, trabecular number and increased trabecular spacing in the spine and femur, and a decrease in cortical thickness in the femur. This closely resembles what is described for tibias from adult OI type I patients as measured with high resolution peripheral quantitative

computed tomography.^{8,38–40} Biomechanical testing shows that both the *Col1a1*^{+/-} model and the Mov13 mouse model show a decrease in post yield displacement, suggesting the bones are more brittle. *Col1a1*^{+/-} mice also show decreased stiffness as well as decrease in ultimate load not seen in the Mov13 mouse,^{20,37} suggesting that *Col1a1*^{+/-} mice have bones that are less resistant to bending and fracture than the Mov13 model. On a cellular level, the Mov13 mouse shows no change in osteoclast number/bone perimeter.²⁰ In OI type I patients, an increase in osteoblast number, osteoclast number and osteoclast surface has been reported.^{28,29,41} Similar to what has been reported in patients, we observed an increase in osteoclast number and a trend towards increased osteoblast number in the *Col1a1*^{+/-} mouse. No significant changes were reported for dynamic histomorphometry in the Mov13 mice³⁷ and dynamic histomorphometry was not reported for the *Col1a1*^{+/-365} mice. We observe a decrease in MAR in the *Col1a1*^{+/-} mice. This is consistent with what is seen in OI type I patients. In children with OI type I, an increase in BFR has been reported,^{28,41} while decreased or unchanged BFR has been reported in adult patients.^{30,31,42} We observe a decrease in BFR in our adult mice, consistent with adult patients. Although direct comparison isn't possible due to difference in timepoints studied, we show comparable changes in our *Col1a1*^{+/-} mouse compared to the *Col1a1*^{+/-365} mouse in both μ CT and biomechanical values. Only osteoclast histomorphometric measurements are available for the *Col1a1*^{+/-365} mouse and these show a similar trend. This is not surprising as the *Col1a1*^{+/-365} mouse also has an exon 2-5 deletion in the *Col1a1* gene.¹⁶

The reason the Mov13 mouse shows different but overlapping phenotype might be because while the viral insertion in Mov13 mice completely limits *Col1a1* expression in fibroblasts, osteoblasts show limited capability of processing the Mov13 allele such that functional *Col1a1* mRNA is produced.¹⁹ Hence, it is not haploinsufficient for type I collagen in bone. In long bone, Mov13 mice have been described to either have an increase¹⁸ or no change²⁰ in *Col1a1* mRNA at 8 wk. We see a reduction in *Col1a1* mRNA and pro-COL1A1 protein in the *Col1a1*^{+/-} bone at 12 wk of age.

One of the limitations of this study is that we did not examine the changes in phenotype over time. The Mov13 mice were shown to recover in bone failure load and bone density by 16 wk. This recovery in strength is likely a result of geometrical remodeling and included an increase in cortical area.¹⁸ This is reminiscent of what is seen in patients where fractures are seen more commonly during childhood. Unlike the Mov13 model, we see persistent reduced bone mass and strength at 16 wk. However, we did not study early changes or changes in aged mice. Future studies would be interesting to compare different ages.

In summary, we describe a novel *Col1a1* haploinsufficient mouse model that presents with low bone mass, bone brittleness and reduced resistance to breaking in 3-point bending. Our results show this mouse responds with an increase in bone mass and strength upon treatment with anti-TGF- β antibody, 1D11. This supports the expanding the study of anti-TGF- β treatment to patients with OI caused by quantitative defects in type I collagen.

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Author contributions

Ellen Busschers (Conceptualization, Formal analysis, Investigation, Writing—original draft, Writing—review & editing), Yuqing Chen-Everson (Formal analysis, Investigation), Mary Adeyeye (Formal analysis, Investigation, Writing—review & editing), I.-Wen Song (Conceptualization, Funding acquisition, Investigation, Writing—review & editing), Emily Busse (Formal analysis, Writing—review & editing), Alexis Castellon (Formal analysis), Oscar Ruiz (Data curation, Formal analysis, Writing—review & editing), Nicole Meyers (Formal analysis, Investigation), Yangjin Bae (Writing—review & editing), Catherine Ambrose (Formal analysis, Writing—review & editing), Brendan Lee (Conceptualization, Funding acquisition, Supervision, Writing—review & editing)

Ellen Busschers and Yuqing Chen-Everson contributed equally to this work.

Supplementary material

Supplementary material is available at *Journal of Bone and Mineral Research* online.

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Conflicts of interest

Disclosure the 1D11 and 13C4 antibodies were provided by Sanofi. Baylor College of Medicine has licensed the discovery "Treatment of Osteogenesis Imperfecta or brittle bone disease with anti-TGF β treatment" to Sanofi.

Data availability

All data analyzed in this study are included in the manuscript and supplementary figures. RNA-seq data discussed in this study are accessible through NCBI's Gene Expression Omnibus (GEO) database by GEO series accession number GSE293448 <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE293448>.

Ethics approval statement

All mouse experiments were performed according to a protocol approved by the Institutional Animal Care and Use Committee at Baylor College of Medicine.

References

1. Jovanovic M, Guterman-Ram G, Marini JC. Osteogenesis imperfecta: mechanisms and signaling pathways connecting classical and rare OI types. *Endocr Rev.* 2022;43(1):61-90. <https://doi.org/10.1210/edrv/bnab017>

2. Rossi V, Lee B, Marom R. Osteogenesis imperfecta: advancements in genetics and treatment. *Curr Opin Pediatr*. 2019;31(6):708-715. <https://doi.org/10.1097/MOP.0000000000000813>
3. Forlino A, Marini JC. Osteogenesis imperfecta. *Lancet*. 2016;387(10028):1657-1671. [https://doi.org/10.1016/S0140-6736\(15\)00728-X](https://doi.org/10.1016/S0140-6736(15)00728-X)
4. Rauch F, Glorieux FH. Osteogenesis imperfecta. *Lancet*. 2004;363(9418):1377-1385. [https://doi.org/10.1016/S0140-6736\(04\)16051-0](https://doi.org/10.1016/S0140-6736(04)16051-0)
5. Silience DO, Senn A, Danks DM. Genetic heterogeneity in osteogenesis imperfecta. *J Med Genet*. 1979;16(2):101-116. <https://doi.org/10.1136/jmg.16.2.101>
6. Ben Amor IM, Roughley P, Glorieux FH, Rauch F. Skeletal clinical characteristics of osteogenesis imperfecta caused by haploinsufficiency mutations in COL1A1. *J Bone Miner Res*. 2013;28(9):2001-2007. <https://doi.org/10.1002/jbmr.1942>
7. Bardai G, Lemyre E, Moffatt P, et al. Osteogenesis imperfecta type I caused by COL1A1 deletions. *Calcif Tissue Int*. 2016;98(1):76-84. <https://doi.org/10.1007/s00223-015-0066-6>
8. Hald JD, Folkestad L, Harsløf T, et al. Skeletal phenotypes in adult patients with osteogenesis imperfecta—correlations with COL1A1/COL1A2 genotype and collagen structure. *Osteoporos Int*. 2016;27(11):3331-3341. <https://doi.org/10.1007/s00198-016-3653-0>
9. Dinulescu A, Păsărică AS, Carp M, et al. New perspectives of therapies in osteogenesis imperfecta—a literature review. *J Clin Med*. 2024;13(4):1065. <https://doi.org/10.3390/jcm13041065>
10. Muñoz-García J, Heymann D, Giurgea I, et al. Pharmacological options in the treatment of osteogenesis imperfecta: a comprehensive review of clinical and potential alternatives. *Biochem Pharmacol*. 2023 Jul;213:115584. <https://doi.org/10.1016/j.bcp.2023.115584>
11. Edwards JR, Nyman JS, Lwin ST, et al. Inhibition of TGF- β signaling by 1D11 antibody treatment increases bone mass and quality in vivo. *J Bone Miner Res*. 2010;25(11):2419-2426. <https://doi.org/10.1002/jbmr.139>
12. Grafe I, Yang T, Alexander S, et al. Excessive transforming growth factor- β signaling is a common mechanism in osteogenesis imperfecta. *Nat Med*. 2014;20(6):670-675. <https://doi.org/10.1038/nm.3544>
13. Greene B, Russo RJ, Dwyer S, et al. Inhibition of TGF- β increases bone volume and strength in a mouse model of osteogenesis imperfecta. *JBM R Plus*. 2021;5(9):e10530. <https://doi.org/10.1002/jbm4.10530>
14. Bi X, Grafe I, Ding H, et al. Correlations between bone mechanical properties and bone composition parameters in mouse models of dominant and recessive osteogenesis imperfecta and the response to anti-TGF- β treatment. *J Bone Miner Res*. 2017;32(2):347-359. <https://doi.org/10.1002/jbmr.2997>
15. Song IW, Nagamani SCS, Nguyen D, et al. Targeting TGF- β for treatment of osteogenesis imperfecta. *J Clin Invest*. 2022;132(7):e152571. <https://doi.org/10.1172/JCI152571>
16. Liu Y, Wang J, Liu S, et al. A novel transgenic murine model with persistently brittle bones simulating osteogenesis imperfecta type I. *Bone*. 2019 Oct;127:646-655. <https://doi.org/10.1016/j.bone.2019.07.021>
17. Bonadio J, Saunders TL, Tsai E, et al. Transgenic mouse model of the mild dominant form of osteogenesis imperfecta. *Proc Natl Acad Sci*. 1990;87(18):7145-7149. <https://doi.org/10.1073/pnas.87.18.7145>
18. Bonadio J, Jepsen KJ, Mansoura MK, Jaenisch R, Kuhn JL, Goldstein SA. A murine skeletal adaptation that significantly increases cortical bone mechanical properties. Implications for human skeletal fragility. *J Clin Invest*. 1993;92(4):1697-1705. <https://doi.org/10.1172/JCI116756>
19. Kratochwil K, Ghaffari-Tabrizi N, Holzinger I, Harbers K. Restricted expression of Mov13 mutant $\alpha 1(I)$ collagen gene in osteoblasts and its consequences for bone development. *Dev Dyn*. 1993;198(4):273-283. <https://doi.org/10.1002/aja.1001980405>
20. Claeys L, Zhytnik L, Wisse LE, et al. Exploration of the skeletal phenotype of the Col1a1+/Mov13 mouse model for haploinsufficient osteogenesis imperfecta type 1. *Front Endocrinol (Lausanne)*. 2023;14:1145125. <https://doi.org/10.3389/fendo.2023.1145125>
21. Chen Y, Yang S, Lovisa S, et al. Type-I collagen produced by distinct fibroblast lineages reveals specific function during embryogenesis and osteogenesis imperfecta. *Nat Commun*. 2021;12(1):7199. <https://doi.org/10.1038/s41467-021-27563-3>
22. Ge SX, Jung D, Yao R. ShinyGO: a graphical gene-set enrichment tool for animals and plants. *Bioinformatics*. 2020;36(8):2628-2629. <https://doi.org/10.1093/bioinformatics/btz931>
23. Luo W, Brouwer C. Pathview: an R/Bioconductor package for pathway-based data integration and visualization. *Bioinformatics*. 2013;29(14):1830-1831. <https://doi.org/10.1093/bioinformatics/btt285>
24. Kanehisa M, Furumichi M, Sato Y, Ishiguro-Watanabe M, Tanabe M. KEGG: integrating viruses and cellular organisms. *Nucleic Acids Res*. 2021;49(D1):D545-D551. <https://doi.org/10.1093/nar/gkaa970>
25. Erben RG. Embedding of bone samples in methylmethacrylate: an improved method suitable for bone histomorphometry, histochemistry, and immunohistochemistry. *J Histochem Cytochem*. 1997;45(2):307-313. <https://doi.org/10.1177/002215549704500215>
26. Shi CG, Zhang Y, Yuan W. Efficacy of bisphosphonates on bone mineral density and fracture rate in patients with osteogenesis imperfecta. *Am J Ther*. 2016;23(3):e894-e904. <https://doi.org/10.1097/MJT.0000000000000236>
27. Hald JD, Evangelou E, Langdahl BL, Ralston SH. Bisphosphonates for the prevention of fractures in osteogenesis imperfecta: meta-analysis of placebo-controlled trials. *J Bone Miner Res*. 2015;30(5):929-933. <https://doi.org/10.1002/jbmr.2410>
28. Rauch F, Travers R, Parfitt AM, Glorieux FH. Static and dynamic bone histomorphometry in children with osteogenesis imperfecta. *Bone*. 2000;26(6):581-589. [https://doi.org/10.1016/S8756-3282\(00\)00269-6](https://doi.org/10.1016/S8756-3282(00)00269-6)
29. Aksornthong S, Patel P, Komarova SV. Osteoclast indices in osteogenesis imperfecta: systematic review and meta-analysis. *JBM R Plus*. 2024;8(11):ziae112. <https://doi.org/10.1093/jbmrlp/ziae112>
30. McCarthy EF, Earnest K, Rossiter K, Shapiro J. Bone histomorphometry in adults with type IA osteogenesis imperfecta. *Clin Orthop Relat Res*. 1997 Mar;336:254-262. <https://doi.org/10.1097/00003086-199703000-00034>
31. Ste-Marie LG, Charhon SA, Edouard C, Chapuy MC, Meunier PJ. Iliac bone histomorphometry in adults and children with osteogenesis imperfecta. *J Clin Pathol*. 1984;37(10):1081-1089. <https://doi.org/10.1136/jcp.37.10.1081>
32. Tauer JT, Abdullah S, Rauch F. Effect of anti-TGF- β treatment in a mouse model of severe osteogenesis imperfecta. *J Bone Miner Res*. 2019;34(2):207-214. <https://doi.org/10.1002/jbmr.3617>
33. Formenti SC, Lee P, Adams S, et al. Focal irradiation and systemic TGF β blockade in metastatic breast cancer. *Clin Cancer Res*. 2018;24(11):2493-2504. <https://doi.org/10.1158/1078-0432.CCR-17-3322>
34. Vincenti F, Fervenza FC, Campbell KN, et al. A phase 2, double-blind, placebo-controlled, randomized study of fresolimumab in patients with steroid-resistant primary focal segmental glomerulosclerosis. *Kidney Int Rep*. 2017;2(5):800-810. <https://doi.org/10.1016/j.ekir.2017.03.011>
35. Rice LM, Padilla CM, McLaughlin SR, et al. Fresolimumab treatment decreases biomarkers and improves clinical symptoms in systemic sclerosis patients. *J Clin Invest*. 2015;125(7):2795-2807. <https://doi.org/10.1172/JCI77958>
36. Morris JC, Tan AR, Olencki TE, et al. Phase I study of GC1008 (fresolimumab): a human anti-transforming growth factor-Beta

- (TGF β) monoclonal antibody in patients with advanced malignant melanoma or renal cell carcinoma. *PLoS One*. 2014;9(3):e90353. <https://doi.org/10.1371/journal.pone.0090353>
37. Jacobsen CM, Schwartz MA, Roberts HJ, et al. Enhanced Wnt signaling improves bone mass and strength, but not brittleness, in the Col1a1 +/-mov13 mouse model of type I osteogenesis imperfecta. *Bone*. 2016 Sep;90:127-132. <https://doi.org/10.1016/j.bone.2016.06.005>
 38. Folkestad L, Hald JD, Hansen S, et al. Bone geometry, density, and microarchitecture in the distal radius and tibia in adults with osteogenesis imperfecta type I assessed by high-resolution pQCT. *J Bone Miner Res*. 2012;27(6):1405-1412. <https://doi.org/10.1002/jbmr.1592>
 39. Rolvien T, Stürznickel J, Schmidt FN, et al. Comparison of bone microarchitecture between adult osteogenesis imperfecta and early-onset osteoporosis. *Calcif Tissue Int*. 2018;103(5):512-521. <https://doi.org/10.1007/s00223-018-0447-8>
 40. Kocijan R, Muschitz C, Haschka J, et al. Bone structure assessed by HR-pQCT, TBS and DXL in adult patients with different types of osteogenesis imperfecta. *Osteoporos Int*. 2015;26(10):2431-2440. <https://doi.org/10.1007/s00198-015-3156-4>
 41. Roschger P, Fratzl-Zelman N, Misof BM, Glorieux FH, Klaushofer K, Rauch F. Evidence that abnormal high bone mineralization in growing children with osteogenesis imperfecta is not associated with specific collagen mutations. *Calcif Tissue Int*. 2008; 82(4):263-270. <https://doi.org/10.1007/s00223-008-9113-x>
 42. Iwamoto J, Takeda T, Ichimura S. Increased bone resorption with decreased activity and increased recruitment of osteoblasts in osteogenesis imperfecta type I. *J Bone Miner Metab*. 2002; 20(3):174-179. <https://doi.org/10.1007/s007740200025>